DP Computer Science: A 2.2.3 Network Models

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Define and differentiate between **client-server** and **peer-to-peer** network models
- Analyse the **advantages and disadvantages** of each model
- Identify **real-world applications** of each model
- Evaluate the **implications** of each model for **data privacy and security**

Key Vocabulary

Term	Definition
Client-server model	A network model where a central server provides services to multiple client devices
Peer-to-peer (P2P) model	A network model where devices share resources directly without a central server
Client	A device that requests services from a server
Server	A device that provides services to clients
Peer	A device in a P2P network that acts as both a client and a server
Scalability	The ability to handle increasing numbers of users or devices
Single point of failure	A component whose failure causes the entire system to fail
Decentralised	Distributed across many devices, with no single point of control
Hybrid model	A combination of client-server and peer-to-peer approaches

IB ATL Skills

Skill	Description
Thinking	Critical thinking, analysis, comparing and contrasting
Communication	Explaining concepts and collaborating in discussions
Self-Management	Organisation, time management
Research	Gathering information and evaluating sources (optional extension)

2. Network Models Overview

What is a Network Model?

> **A network model is a framework that describes how computers communicate and share resources.**

There are two main models:

Model	Key Feature	Example
Client-Server	Central server provides services to many clients	Web browsing, email, online banking
Peer-to-Peer (P2P)	Devices share resources directly with no central server	File sharing (BitTorrent), blockchain

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Visual Comparison

Client-Server Model:

```text

  ┌───────────────────────────────────────────────────────┐
  │                   CLIENT-SERVER MODEL                  │
  ├───────────────────────────────────────────────────────┤
  │                                                       │
  │   ┌───────────────────────────────────────────────┐   │
  │   │                    SERVER                      │   │
  │   │          • Stores data centrally              │   │
  │   │          • Provides services to all clients   │   │
  │   │          • Manages security and access        │   │
  │   └───────────────────────────────────────────────┘   │
  │                         │                             │
  │                         │                             │
  │          ┌──────────────┼──────────────┐              │
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  │          ▼              ▼              ▼              │
  │      ┌────────┐     ┌────────┐     ┌────────┐         │
  │      │ Client │     │ Client │     │ Client │         │
  │      │  (PC)  │     │(Laptop)│     │(Phone) │         │
  │      │Requests│     │Requests│     │Requests│         │
  │      └────────┘     └────────┘     └────────┘         │
  │                                                       │
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```

Peer-to-Peer Model:

```text

  ┌───────────────────────────────────────────────────────┐
  │                   PEER-TO-PEER MODEL                  │
  ├───────────────────────────────────────────────────────┤
  │                                                       │
  │             ┌─────────────────────────┐               │
  │             │     All Peers Share     │               │
  │             │   Resources Directly    │               │
  │             └─────────────────────────┘               │
  │                                                       │
  │                  ┌────────┐                           │
  │             ┌───▶│  Peer  │◀───┐                      │
  │             │    │  (PC)  │    │                      │
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│         │    └─────┘    │                   │
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│         │       │       │                   │
│     ┌───┴──┐    │    ┌──┴───┐               │
│     │ Peer │────┴────│ Peer  │              │
│     │(Laptop)│  │    │ (Phone) │            │
│     └──────┘    │    └──────┘               │
│                 │                           │
│              ┌──┴───┐                       │
│              │ Peer │                       │
│              │(Tablet)│                     │
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3. Client-Server Model

Definition

> **A network model where a central server provides services to multiple client devices. Clients request services; the server provides them.**

How It Works

1. **Clients** request services (files, web pages, email)
2. **Server** processes requests and sends responses
3. Server manages **authentication** and **access control**
4. Data is stored **centrally** on the server

Advantages

Advantage	Description
Centralised Management	Easy to manage users, files, and permissions from one place
Scalability	Can add more clients without changing the server (up to a point)
Data Backup	All data stored centrally—easier to back up
Security	Centralised security controls (firewalls, authentication)
Reliability	Professional servers are built for continuous operation
Consistent Data	Everyone accesses the same version of data

Disadvantages

Disadvantage	Description
Single Point of Failure	If the server fails, the whole network goes down
Cost	Servers and server software are expensive
Bottleneck	All requests go through the server; can become overwhelmed

Disadvantage	Description
Maintenance	Requires skilled IT staff to manage
Centralised Control	All data is in one place—potentially a target for attackers

Real-World Applications

Application	Why Client-Server?
Web Browsing	Websites are hosted on web servers; browsers are clients
Email	Email is sent and received via mail servers
Online Banking	Central server manages all accounts and transactions
School Network	Central server manages student accounts, files, and printers
Corporate Systems	Employees access shared resources via central servers

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4. Peer-to-Peer (P2P) Model

Definition

> **A network model where devices (peers) share resources directly with each other without a central server. Every device can act as both a client and a server.**

How It Works

1. Each **peer** can provide and request resources
2. No **central server** manages the network
3. Peers communicate **directly** with each other
4. Resources are **distributed** across all peers

Advantages

Advantage	Description
No Single Point of Failure	No central server to fail—network remains functional
Cost-Effective	No expensive server hardware required
Scalable	More peers = more resources available
Decentralised	No central authority; resilient
Resilience	Even if some peers leave, others remain available
Direct Communication	Can be faster for nearby peers

Disadvantages

Disadvantage	Description
Security	No central security management; harder to control access
Management Challenges	No central administration; harder to manage
Performance Variations	Depends on peer availability and bandwidth
Data Availability	Files may not be available if peers leave
Legal Issues	Often associated with copyright infringement
Trust Issues	No central authority to verify peers

Real-World Applications

Application	Why P2P?
BitTorrent	Files are downloaded from multiple peers simultaneously
Blockchain	Distributed ledger; each peer has a copy of the data
Cryptocurrency	Bitcoin is a P2P network; no central bank
Skype (Legacy)	Used P2P for direct voice/video calls (now mostly client-server)
File Sharing	Sharing files directly between users (Napster, LimeWire)

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5. Comparison Table

Feature	Client-Server	Peer-to-Peer
Central Authority	Yes (server)	No (decentralised)
Data Storage	Centralised on server	Distributed across peers
Security	Centralised; can be tightly controlled	Decentralised; harder to secure
Scalability	Limited by server capacity	Highly scalable (more peers = more resources)
Single Point of Failure	Yes (server)	No
Cost	High (server hardware/software)	Low (no server required)
Management	Centralised; easy to manage	Decentralised; complex
Performance	Consistent (with sufficient server power)	Variable (depends on peers)
Data Backup	Easy (centralised)	Harder (distributed)
Examples	Web browsing, email, banking	BitTorrent, blockchain, cryptocurrency

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6. Security and Privacy Implications

Aspect	Client-Server	Peer-to-Peer
Data Privacy	Server has access to all data; must be trusted	Data is distributed; less centralised collection
Security Controls	Centralised; can enforce strong security policies	Decentralised; harder to enforce security
Vulnerability	Attackers target the server	Attackers may target peers individually
Data Ownership	Server provider controls data	Users have more control over their data
Encryption	Often implemented centrally (HTTPS, TLS)	May be implemented but less consistent
Access Control	Centralised user authentication	Peer-based authentication

Key Privacy Consideration:

Model	Privacy Implication
Client-Server	The server provider can access all data. Trust is required in the service provider.
Peer-to-Peer	Data is distributed. No central point can access all data.

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7. Hybrid Models

What is a Hybrid Model?

> **Many modern systems use a combination of client-server and peer-to-peer approaches—taking the best features of both.**

Example: Messaging Apps

Feature	Model Used	Why?
User Authentication	Client-Server	Centralised user management
Contact Lists	Client-Server	Stored centrally; accessible from any device
Message History	Client-Server	Stored centrally for access across devices
Real-Time Calls	Peer-to-Peer (or P2P hybrid)	Direct connection reduces latency and server load

Common Hybrid Examples

Application	Hybrid Approach
Skype	Uses central servers for authentication, but direct P2P for media transfer

Application	Hybrid Approach
Spotify	Central server for streaming; P2P used for some content delivery
WhatsApp	Central server for messages; end-to-end encryption ensures privacy
Blockchain Apps	P2P for the blockchain; some use central services for user interfaces

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8. Scenario-Based Activity

Instructions

For each scenario, determine which network model is most suitable and justify your choice.

Scenario 1: School Network

Problem: A school network needs to provide students and teachers access to shared files, printers, and the internet.

Feature Required	Model Preferred	Why?
User Accounts	Client-Server	Centralised user management
File Access	Client-Server	Controlled file access; easier to secure
Printers	Client-Server	Centralised print management
Internet Access	Client-Server	Can filter and monitor usage centrally
Scalability	Client-Server	Supports many users

Most Suitable Model: Client-Server

Why: Centralised management of resources, user accounts, and security. Easier to back up data and control access.

Scenario 2: Online Gaming

Problem: A group of friends wants to play an online game together.

Game Type	Preferred Model	Why?
Massive Multiplayer	Client-Server	Central server manages game state; ensures fair play
Small Private Game	P2P	Direct connection; lower latency; reduced costs

Most Suitable Model: Can be either, depending on the game.

- **Client-Server:** Central server manages game state, player interactions, and ensures fair play.
- **P2P:** Direct communication between players; good for small groups.

Scenario 3: Messaging App

Problem: A messaging app needs to allow users to send text messages, images, and make voice and video calls.

Feature	Model Used	Why?
Authentication	Client-Server	Centralised user management
Contact Lists	Client-Server	Accessible from any device
Text Messages	Client-Server	Reliable delivery; stored centrally
Voice/Video Calls	P2P (or hybrid)	Direct connection improves quality; reduces server load

Most Suitable Model: Hybrid (Client-Server + P2P)

Why: Client-server for authentication and message delivery; P2P for real-time communication.

Scenario 4: Community Library

Problem: A library wants to create a system for sharing e-books among its members.

Feature Required	Model Preferred	Why?
Central Catalogue	Client-Server	All books listed in one place
Borrowing Management	Client-Server	Track who has borrowed what
User Accounts	Client-Server	Manage members and their borrowing history
Access Control	Client-Server	Control who can read which books

Most Suitable Model: Client-Server

Why: Centralised management of the e-book collection, user accounts, and borrowing privileges. Easier to track availability and implement access controls.

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9. Lesson Sequence

Lesson Plan

Phase	Time	Activity
Engage	10 min	"How does your phone get its music from Spotify?"; introduce network models
Explore	25 min	Explore client-server (web browsing, email) and P2P (file sharing, blockchain) models
Interactive Activity	15 min	Think-Pair-Share: match scenarios to models and justify choices
Wrap-up	10 min	Comparison of models; security and privacy implications; exit ticket

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Activity: Comparison Table

Instructions: Complete the table comparing client-server and peer-to-peer models.

Feature	Client-Server	Peer-to-Peer
Central Authority	Yes (server)	
Data Storage	Centralised	
Security	Centralised	
Single Point of Failure	Yes	
Cost	High	
Management	Easy	
Examples		BitTorrent, blockchain

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10. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Comparison Table	Complete table comparing models
Scenario Analysis	Recommend suitable model and justify choices
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is the key difference between client-server and peer-to-peer networks?
2. What is the main disadvantage of a client-server network?
3. Give one example of a peer-to-peer application.
4. Which model is more secure? Why?

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11. Differentiation

Level	Strategy
Support	Provide visual aids, concise definitions, and real-world examples
Challenge	Encourage advanced students to research hybrid models or explore blockchain technology in more depth

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12. Resources

- **Presentation** (Slide Deck)
- **eBook** (A2.2.3 with quiz)
- **Worksheet:** Comparison table + scenario analysis
- **Quizlet** (A2.2.3 vocabulary flashcards)

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13. Summary: Key Takeaways

Model	Key Feature	Advantages	Disa
Client-Server	Central server provides services	Centralised management, scalable, secure	Sing
Peer-to-Peer	Devices share directly	No single point of failure, cost-effective, resilient	Secu
Hybrid	Combination of both	Flexible, balances strengths/weaknesses	Com

> *"The choice of network model depends on the specific needs—client-server offers control and security; peer-to-peer offers resilience and cost-effectiveness."*